Project | Sustainability Impact Analysis for Intel

Key Insights

1	Total number of devices Intel repurposed in 2024	19719
2	Average age of repurposed devices in 2024	3.9
3	Average estimated energy savings (kWh) from repurposed devices per year compared to new devices	28.4
4	Total CO2 emissions saved (in tons) from repurposed devices	276.2

With one tree absorbing about 0.05 tons of CO2 a year, the savings with the repurposed devices is equivalent to plnting over 5,500 trees, connecting the long-term carbon offset efforts.

Patterns and Trends

1	Total repurposed laptops	16512
2	Total repurposed desktops	3207
3	Average energy savings for repurposing laptops	28.37
4	Average energy savings for repurposing desktops?	28.77

Energy Savings by Device Age

Device Age	# of Devices	Average Energy Saved (kWh)
1	1003	12.1
2	1962	16.6
3	5683	23.8
4	4630	28.3
5	3595	35.1
6	1706	40.0
7	785	46.6
8	355	52.5

Most of the repurposed devices are between the ages of 3-5 years, but a trend to follow is the older the device, the more energy is saved.

Reccomendation for Intel

To maximize energy savings and C02 reductions, Intel should prioritize the repurposing of devices that are 3-5 years old, with a strategic emphasis on laptops, which are the most commonly repurposed device and collectively contribute the most to overall impact. They should develop targeted collection and outreach campaigns focusing on recovering these devices from corporate partners, schools, and consumers as these devices have the best balance between energy efficieny and availability.